

Gridlabd: Open Source Software for Smart Grid Network Studies

The distribution of electric supply generally faces a few problems. The good news is that these problems can be modelled mathematically and studied without any danger to life and limb. Gridlabd is open source software which enables smart grid studies.



Economising the operation of electricity generators is a problem (called the 'economic dispatch problem') that has been faced since the 18th century. Mathematical models have been developed to study such problems. Another problem is the transmission of power from the generator to the customer, which is called 'optimal power flow'. The third is the 'unit commitment problem', which refers to finding the least-cost ON/OFF state of the available power generation resources to meet the electrical load.

Economising power system operations is still a problem; hence, the term 'smart grid'. A smart-grid is capable of modelling and studying each device connected to the distribution system, which is highly inefficient compared to generation and transmission systems. Therein lies the importance of Gridlabd software. Gridlabd is capable of modelling each and every component in the power system with mathematics running in its core. It is possible to simulate time-series characteristics from micro-seconds to years. Hence, various cases can be studied on a computer before actual system implementation. It helps in improving the efficiency of an existing system and the future expansion of the distribution system. Figure 1 (courtesy Wikipedia) shows a typical power system network.

Gridlabd (GLD)

Gridlabd was developed by the US Department of Energy

(DOE) at Pacific Northwest National Laboratory (PNNL) under funding from the Office of Electricity. While writing this article, the source code for Gridlabd was not available in Ubuntu (am presuming you're a Debian user). So the available Red Hat package manager (rpm) has been converted to Debian (Deb). Gridlabd is the first simulation platform in which modern energy systems are inbuilt. Time series simulation and load modelling become easier with Gridlabd from sub-station to customers (residential load). Gridlabd is an open source tool freely available to anyone. It encourages collaboration with industry and academia. The BSD-style licence allows us to add or extract our own modules without compromising the internal intellectual property. The Web portal <http://www.gridlabd.org/gldportal> has a GUI for quick real-time hands-on operation of Gridlabd.

Installation and how it works

Since Gridlabd is not readily available for installation in Ubuntu, we need to convert the available rpm package to Deb. The steps to be followed are given here.

Open a terminal and download the package

```
hithu@linux:~$ wget https://sourceforge.net/projects/
gridlab-d/files/gridlab-d/Last%20stable%20release/
gridlabd-3.2.0-1.x86_64.rpm.
```